

Parameterized Cancellation of Partial-Band Partial-Block-Duration Interference for Underwater Acoustic OFDM

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Abstract—Despite that underwater acoustic channels are well known to contain various interferences, research on interference mitigation in underwater acoustic communications has been very limited. In this paper, we deal with a wideband orthogonal frequency division multiplexing (OFDM) transmission in the presence of an external interference which occupies partially the signal band and whose time duration is shorter than the OFDM block. We parameterize the unknown interference waveform by a number of parameters assuming prior knowledge of the frequency band and time duration of the interference, and develop an iterative receiver, which couples interference detection via a generalized likelihood-ratio-test (GLRT), interference reconstruction and cancellation, channel estimation, and data detection. In addition to simulation results, we verify the receiver performance using data sets collected from two experiments. In both time-invariant and time-varying channels, the proposed iterative receiver achieves robust performance in the presence of unknown interference.

Index Terms—Orthogonal frequency division multiplexing (OFDM), parameterized iterative interference mitigation, partial-band short-duration interference, underwater acoustic communications.

I. INTRODUCTION

INTERFERENCE mitigation in communication systems has been an interesting topic since the advent of communication. As opposed to the internal interference, such as the intersymbol interference (ISI) in single carrier transmissions, or intercarrier interference (ICI) in multicarrier transmissions, interference from the external environment, such as jamming and unintentional interference, could result in significant performance degradation. The interference in underwater acoustic (UWA) communications includes both unintentional interference, such as interference from sonar operations and marine mammals, and malicious jamming [1].

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This work was primarily motivated by our observations in an experiment held in March 2010, over one underwater network deployed at the Atlantic Undersea Test and Evaluation Center (AUTEC) around Andros Island near the Tongue of the Ocean, Bahamas [2], [3]. During the transmission of communication signals, other unknown users were transmitting multiple sonar waveforms in the same environment. The received communication signal at certain sensors was thus superimposed with the sonar transmissions as shown in Fig. 1, which leads to significant performance degradation. Although such interference occurs quite often during the UWA communications, limited research results for its mitigation are available. For example, although considerable advancements have been made for orthogonal frequency division multiplexing (OFDM) and single-carrier transmissions with frequency domain equalization in recent years; see, e.g., [4]–[16], and references therein, all of these are investigated in the absence of interference.

On the other hand, various methods for interference suppression have been investigated for wireless communications in the last two decades [17], [18]. Mainly, the external interferences are divided into two categories according to the time-frequency characteristics: i) impulsive interference with short time duration and large bandwidth; and ii) narrowband interference with small bandwidth and long time duration.

A. Impulsive Noise Suppression

For impulsive noise suppression in multicarrier transmissions, the characteristics of short time duration is usually utilized. Using the time-domain clipping, an erasure/Reed-Solomon (RS) decoding scheme for impulsive noise mitigation was proposed in [19]; the proposed scheme was improved in [20] by performing joint erasure marking and Viterbi decoding, but at the cost of high computational complexity. An iterative interference mitigation strategy was proposed in [21], where the frequency components of the impulsive noise and the information symbols were estimated iteratively, and a threshold detector was used to reconstruct the impulsive noise events in the time domain. The proposed receiver structure was further developed in [22] to increase its convergence rate by employing a syndrome decoder. However, both approaches assume perfect channel knowledge, which is impractical in most cases. For channel estimation in the presence of impulsive noise, a support vector machine based algorithm was proposed in [23] by introducing a ϵ -Huber cost function. An iterative channel estimator was developed in [24] by performing impulsive noise estimation/cancellation and channel estimation iteratively.

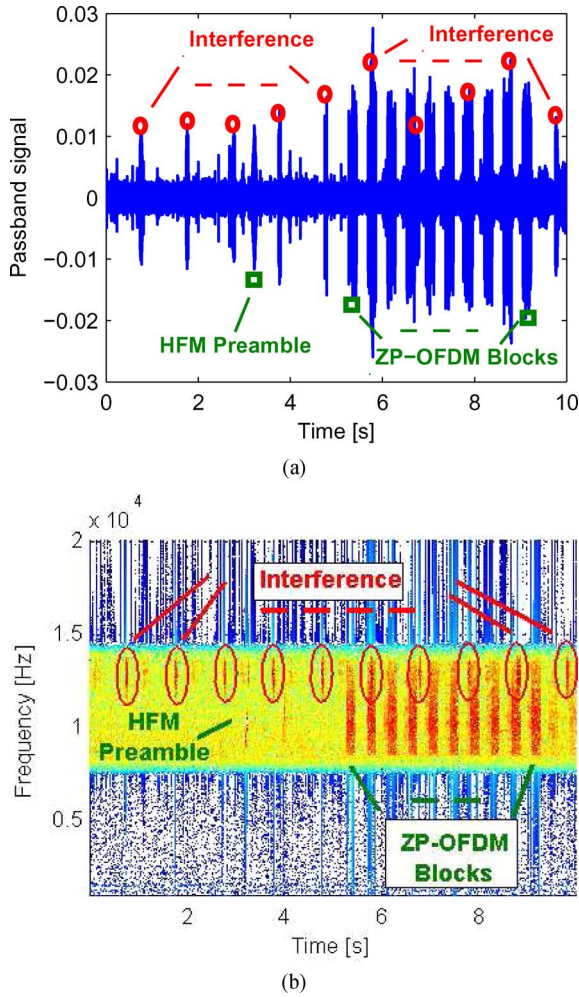


Fig. 1. Samples of the time domain waveform and the time-frequency spectrum of the received signal at one hydrophone. The transmitted signal consists of a hyperbolic frequency modulated (HFM) preamble followed by 10 zero-padded (ZP)-OFDM blocks. The circle and the square in (a) denotes the locations of interference and useful signal, respectively. (a) Time domain waveform after bandpass filtering. (b) Time-frequency spectrum.

B. Narrowband Interference Suppression

For narrowband interference suppression, the frequency characteristics of the interference are usually required. In [25], a linear minimum mean-square-error (LMMSE) estimate of the interference across multiple OFDM blocks is obtained based on a few measurements near the obstructed frequency band. However, this method requires not only the prior knowledge of the center frequency and bandwidth of the interference, but also its power spectral density. The method in [26], [27] performs a successive estimation of the information symbol at one subcarrier by subtracting the interference component predicted from the previous subcarrier, followed by the prediction of the interference at the next subcarrier. In [28], an expectation-maximization (EM)-based iterative channel estimation and symbol decoding scheme in the frequency domain was proposed for a single-user and multi-antenna OFDM system with synchronous interferences, where both spatial and frequency correlation of interference measurements are exploited. Another EM-based iterative channel estimation and data decoding algorithm was proposed

in [29], [30]. By assuming that the variance of the interference follows an inverse-Gamma distribution, the unknown variance of the interference at each subcarrier is treated as a nuisance parameter which can be averaged out. Several other interference estimation or detection schemes can be also found in, e.g., [31]–[33]. In [31], the code structure was exploited for interference detection in the OFDM-based cognitive system. A compressive-sensing based interference localization algorithm was developed in [32] by assuming perfect channel knowledge. Based on message-passing techniques, a factor-graph based interference cancellation and information symbol estimation framework was developed in [33].

C. Our Work

We consider interference mitigation for OFDM transmissions in underwater acoustic channels. Relative to the large signal bandwidth (e.g., 5 ~ 10 kHz) and long block duration (e.g., 100 ~ 200 ms), the interference usually occupies part of the signal band and has shorter time duration. As such, the interference could neither be treated as impulse noise nor narrowband interference, as illustrated in Fig. 1. Conventional interference mitigation methods which exploit either short time duration or narrowband property may not work well for this type of interference. Specifically, for the impulsive noise mitigation methods, the nonlinear operation, such as the clipping and nulling operations, could lead to considerable information loss when the time duration of interference is not relatively small; for the narrowband interference mitigation methods, the assumption that interference is independent across subcarriers might not be valid, which could result in considerable performance degradation in the low SNR scenario, especially in a time-varying scenario where ICI exists.

For such a type of interference with short time duration and partial bandwidth, we represent its waveform by a number of parameters by assuming prior knowledge of the frequency band and time duration of the interference, where the number of parameters is equal to the time-bandwidth product. After incorporating the interference representation into the system model [7], [13], a receiver framework is proposed to perform interference detection, estimation, and data decoding iteratively.

The main contributions of this work are the following.

- We provide a parameterized representation of the interference explicitly such that interference cancellation can be carried out without information loss nor performance degradation caused by approximations.
- We develop an iterative OFDM receiver, which couples interference detection, interference reconstruction and cancellation, channel estimation, and data detection.
- Besides extensive simulation results, we provide validation based data sets collected in two experiments. Both time-invariant and time-varying scenarios have been considered.

Remark 1: In this paper, we propose a systematic approach to mitigating short-time duration partial-band interference in underwater acoustic communications. Note that this approach is quite general that the interference is parameterized by a number of unknowns depending on the time-bandwidth product. With a given time-bandwidth product, the time duration or the bandwidth of the interference could be arbitrary. Hence, the proposed

interference cancellation receiver could also be applicable to the mitigation of impulsive noise and narrowband interference; instead of only utilizing some properties of impulsive noise and narrowband interference to suppress them, one could directly estimate the interference through suitable parameterization and pursue joint or iterative interference cancellation and data detection. We are currently pursuing this approach for the mitigation of impulsive noise, where comparison will be carried out with respect to existing methods, e.g., in [20], [22], [34]–[36].

Remark 2: The proposed approach relies on the prior knowledge of the interference frequency band and waveform duration. These parameters could be acquired based on the received waveforms in the absence of useful signals, as in the operation situation of the AUTECH network. The focus of this work is on unintentional interferences. For malicious interferences with fast-varying waveform characteristics, further research is needed.

The rest of the paper is organized as follows. The system model is introduced in Section II. The interference model is presented in Section III. The proposed interference cancellation receiver is developed in Section IV, with detailed descriptions on several key modules. Numerical simulations are presented in Section V, and experimental results are provided in Sections VI and VII. We draw conclusions in Section VIII.

Notation: Bold upper case letters and lower case letters are used to denote matrices and column vectors, respectively. $(\cdot)^T$ and $(\cdot)^H$ denote transpose and Hermitian transpose, respectively. $\propto$ denotes equality of functions up to a scaling factor. $[\mathbf{a}]_m$ denotes the m th element of vector $\mathbf{a}$, and $[\mathbf{A}]_{m,k}$ denotes the (m,k) th element of matrix $\mathbf{A}$. $\mathbf{I}_N$ denotes an identity matrix of size $N \times N$.

II. SYSTEM MODEL

A. Transmitted Signal and Channel Model

Let T denote the OFDM symbol duration and T_g the length of the guard interval between consecutive OFDM blocks, which leads to the OFDM block duration $T' = T + T_g$. With the subcarrier spacing of $\frac{1}{T}$, a total of K subcarriers are located at frequencies

$$f_k = f_c + \frac{k}{T}, \quad k = -\frac{K}{2}, \dots, \frac{K}{2} - 1 \quad (1)$$

where f_c is the center frequency, and K is assumed even. The signal bandwidth is thus $B = \frac{K}{T}$. Let $s[k]$ denote the information symbol on the k th subcarrier, and define $\mathcal{S}_A$ and $\mathcal{S}_N$ as the nonoverlapping sets of active and null subcarriers, respectively, which satisfy $\mathcal{S}_A \cup \mathcal{S}_N = \{-\frac{K}{2}, \dots, \frac{K}{2} - 1\}$. The passband signal of one OFDM block can be expressed as

$$\tilde{s}(t) = 2\text{Re} \left(\sum_{k \in \mathcal{S}_A} s[k] e^{j2\pi f_k t} g(t) \right), \quad t \in [0, T'] \quad (2)$$

where $g(t)$ is a pulse shaping filter with the Fourier transform denoted by $G(f)$.

Assume that the channel consists of N_{pa} discrete paths. The channel impulse response is

$$h(\tau; t) = \sum_{p=1}^{N_{pa}} A_p(t) \delta(\tau - \tau_p(t)) \quad (3)$$

where $A_p(t)$ and $\tau_p(t)$ are the amplitude and delay of the p th path, respectively. Within one OFDM block, we assume that i) the amplitude does not change $A_p(t) \approx A_p$, and ii) the path delay can be approximated as

$$\tau_p(t) \approx \tau_p - a_p t$$

where τ_p is the initial delay, and a_p is the Doppler rate of the p th path, which can be expressed as $a_p = \frac{v_p}{c}$ with v_p being the relative speed of the transmitter and the receiver projected on the p th path, and c is the sound speed in water. The channel impulse response can be reformulated as

$$h(\tau; t) = \sum_{p=1}^{N_{pa}} A_p \delta(\tau - (\tau_p - a_p t)). \quad (4)$$

B. Received Signal Without Interference

Using the path-based channel model, the received signal in the absence of interference is

$$\tilde{y}(t) = \sum_{p=1}^{N_{pa}} A_p \tilde{s}((1 + a_p)t - \tau_p) + \tilde{n}(t) \quad (5)$$

where $\tilde{n}(t)$ is the additive noise.

The following processing is applied on the received signal. First, the main Doppler effect of the received signal is removed through the resampling operation with a resampling factor $(1 + \hat{a})$, leading to the resampled signal as $\tilde{z}(t) = \tilde{y}\left(\frac{t}{(1 + \hat{a})}\right)$ [7]; second, after passband to baseband downshifting, the carrier frequency offset ϵ caused by the residual Doppler effect is compensated through multiplying $e^{-j2\pi\epsilon t}$ with the baseband signal $\tilde{z}(t)$. In this paper, we use the method in [37] to estimate $\hat{a}$ based on a cyclic-prefix (CP)-OFDM preamble; other research on the resampling factor estimation can be found in, e.g., [38]. We use the null-subcarrier based method [39] to estimate $\hat{\epsilon}$, as done in [7].

The frequency measurement at the m th subcarrier is then obtained via the integration

$$\begin{aligned} z[m] &= \frac{1}{T} \int_0^{T+T_g} z(t) e^{-j2\pi(\frac{m}{T} + \hat{\epsilon})t} dt \\ &= \frac{1}{T} \int_0^{T+T_g} \tilde{z}(t) e^{-j2\pi(f_m + \hat{\epsilon})t} dt. \end{aligned} \quad (6)$$

Substituting (2) and (5) into (6) yields

$$z[m] = \sum_{k=-\frac{K}{2}}^{\frac{K}{2}-1} H[m, k] s[k] + w[m] \quad (7)$$

where $w[m]$ is the additive noise sample, and

$$H[m, k] = \sum_{p=1}^{N_{pa}} A'_p e^{-j2\pi(f_m + \hat{\epsilon})\tau'_p} \varrho_{m,k} \quad (8)$$

with

$$b_p = \frac{a_p - \hat{a}}{1 + \hat{a}}, \quad A'_p = \frac{A_p}{1 + b_p}, \quad \tau'_p = \frac{\tau_p}{1 + b_p}$$

$$\varrho_{m,k} = G\left(f_m - f_k + \frac{\hat{\epsilon} - b_p f_m}{1 + b_p}\right).$$

To reduce the computational load in data decoding, the channel matrix is usually assumed to be banded, i.e., only the main diagonal and several off-diagonals on each side are nonzeros

$$\varrho_{m,k} = \begin{cases} G\left(f_m - f_k + \frac{\hat{\epsilon} - b_p f_m}{1 + b_p}\right), & |m - k| \leq D \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

where we term D as the ICI depth of the channel matrix.

Stack transmitted symbols and frequency measurements at all subcarriers into vectors $\mathbf{s}$ and $\mathbf{z}$, respectively,

$$\mathbf{s} = [s[-\frac{K}{2}] \quad s[-\frac{K}{2} + 1] \quad \cdots \quad s[\frac{K}{2} - 1]]^T \quad (10)$$

$$\mathbf{z} = [z[-\frac{K}{2}] \quad z[-\frac{K}{2} + 1] \quad \cdots \quad z[\frac{K}{2} - 1]]^T \quad (11)$$

and collect channel coefficients $H[m, k]$ into a matrix $\mathbf{H}$

$$\mathbf{H} = \begin{bmatrix} H[-\frac{K}{2}, -\frac{K}{2}] & \cdots & H[-\frac{K}{2}, \frac{K}{2} - 1] \\ \vdots & \ddots & \vdots \\ H[\frac{K}{2} - 1, -\frac{K}{2}] & \cdots & H[\frac{K}{2} - 1, \frac{K}{2} - 1] \end{bmatrix}. \quad (12)$$

The frequency-domain measurements are then compactly expressed as

$$\mathbf{z} = \mathbf{H}\mathbf{s} + \mathbf{w} \quad (13)$$

where $\mathbf{w}$ is similar defined as $\mathbf{z}$.

III. INTERFERENCE PARAMETERIZATION

In conventional interference suppression methods, the interference is described either as impulsive noise with large bandwidth or narrowband noise with long time duration. In this section, by assuming prior knowledge of the frequency band and time duration of the interference, we develop a parameterized interference model to address the interference explicitly.

Denote the center frequency, bandwidth, and time duration of the interference as f_{Ic} , B_I , and T_I , respectively. The bandwidth B_I is taken as the 3-dB bandwidth of the interference, however, the knowledge of B_I does not need to very accurate; a loose upperbound on B_I is sufficient.

Let $\tilde{I}(t)$ denote the passband waveform of the interference, and $I(t)$ the baseband waveform. Since $I(t)$ is time limited, it adopts a Fourier-series representation as

$$I(t) = \sum_{l=-\infty}^{\infty} c_l e^{j2\pi \frac{l}{T_I} t}, \quad t \in [0, T_I] \quad (14)$$

where c_l is the coefficient on the basis $e^{j2\pi \frac{l}{T_I} t}$. Without loss of generality, assume that $N_I = \lceil B_I T_I \rceil$ is an even number. Since $I(t)$ is bandwidth-limited to $[-\frac{B_I}{2}, \frac{B_I}{2}]$, the coefficient

c_l is approximately zero for $l < -\frac{N_I}{2}$ or $l \geq \frac{N_I}{2}$. Hence, we rewrite (14) as

$$I(t) \approx \sum_{l=-\frac{N_I}{2}}^{\frac{N_I}{2}-1} c_l e^{j2\pi \frac{l}{T_I} t}, \quad t \in [0, T_I]. \quad (15)$$

The corresponding passband signal is

$$\tilde{I}(t) = 2\text{Re} \left(\sum_{l=-\frac{N_I}{2}}^{\frac{N_I}{2}-1} c_l e^{j2\pi \tilde{f}_l t} \right), \quad t \in [0, T_I] \quad (16)$$

where $\tilde{f}_l := f_{Ic} + \frac{l}{T_I}$. The Fourier transform of $\tilde{I}(t)$ in the frequency band $\mathcal{B}_I := [f_{Ic} - \frac{B_I}{2}, f_{Ic} + \frac{B_I}{2}]$ can be expressed as

$$\tilde{I}(f) = \sum_{l=-\frac{N_I}{2}}^{\frac{N_I}{2}-1} c_l \frac{\sin(\pi(f - \tilde{f}_l)T_I)}{\pi(f - \tilde{f}_l)} e^{-j\pi(f - \tilde{f}_l)T_I}, \quad \forall f \in \mathcal{B}_I. \quad (17)$$

Note that the interference overlaps the received OFDM signal with an unknown delay. Define τ'_I as the delay of the interference relative to the starting point of the OFDM block in which it resides. After the preprocessing of the OFDM receiver, the interference component at the m th subcarrier is

$$\nu[m] = \frac{1}{T} \int_0^{T+T_g} \tilde{I}\left(\frac{t - \tau'_I}{1 + \hat{a}}\right) e^{-j2\pi(f_m + \hat{\epsilon})t} dt. \quad (18)$$

Following the derivation to (17), we can formulate $\nu[m]$ as

$$\nu[m] = e^{-j2\pi \frac{m}{T} \tau_I} \sum_{l=-\frac{N_I}{2}}^{\frac{N_I}{2}-1} \rho_{m,l} u_l \quad (19)$$

where

$$u_l = \frac{(1 + \hat{a})T_I}{T} e^{-j2\pi(f_c + \hat{\epsilon})\tau_I} c_l, \quad \tau_I = \frac{\tau'_I}{1 + \hat{a}},$$

$$\rho_{m,l} = \frac{\sin(\pi((1 + \hat{a})(f_m + \hat{\epsilon}) - \tilde{f}_l)T_I)}{\pi((1 + \hat{a})(f_m + \hat{\epsilon}) - \tilde{f}_l)T_I} e^{-j\pi((1 + \hat{a})(f_m + \hat{\epsilon}) - \tilde{f}_l)T_I}.$$

Stacking interference components at all subcarriers into a vector $\boldsymbol{\nu}$ yields

$$\boldsymbol{\nu} = \boldsymbol{\Lambda}(\tau_I) \boldsymbol{\Gamma}_I \mathbf{u} \quad (20)$$

where $\boldsymbol{\Lambda}(\tau_I)$ is a $K \times K$ diagonal matrix, $\boldsymbol{\Gamma}_I$ is a $K \times N_I$ matrix, and $\mathbf{u}$ is an $N_I \times 1$ vector

$$[\boldsymbol{\Lambda}(\tau_I)]_{m,m} = e^{-j2\pi \frac{m}{T} \tau_I}, \quad [\boldsymbol{\Gamma}_I]_{m,l} = \rho_{m,l},$$

$$\mathbf{u} = \left[u_{-\frac{N_I}{2}}, \dots, u_{\frac{N_I}{2}-1} \right]^T. \quad (21)$$

The received signal in the presence of interference is then formulated as

$$\mathbf{z} = \mathbf{H}\mathbf{s} + \boldsymbol{\Lambda}(\tau_I) \boldsymbol{\Gamma}_I \mathbf{u} + \mathbf{w}. \quad (22)$$

By modeling the interference explicitly, an effective OFDM receiver will be developed in Section IV through iterative estimation of the interference and information symbols.

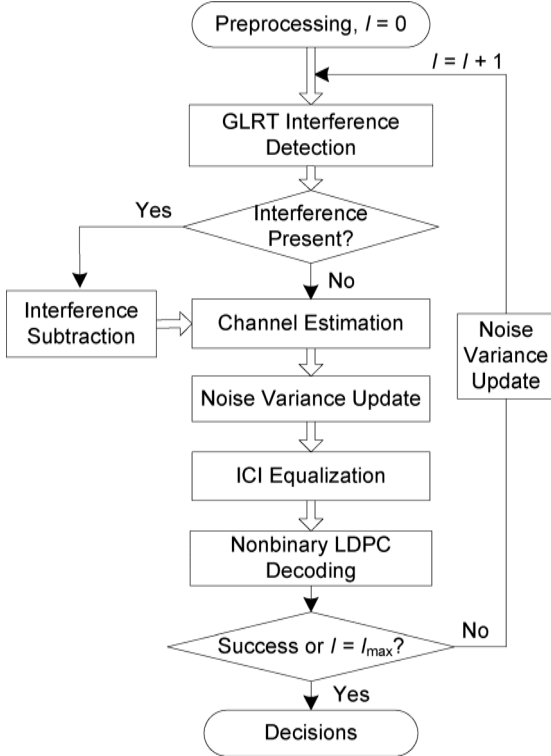


Fig. 2. Receiver structure for interference mitigation and data detection.

IV. PROPOSED OFDM RECEIVER WITH INTERFERENCE CANCELLATION

The received signal model in (22) shows that four sets of parameters: i) the channel mixing matrix $\mathbf{H}$; ii) the information symbol vector $\mathbf{s}$; iii) the interference vector $\mathbf{u}$; and iv) the delay τ_1 , need to be estimated from the same measurements $\mathbf{z}$. Although pilots are usually multiplexed with information symbols to separate channel estimation and information symbol detection in the conventional OFDM receiver [7], frequency measurements at both pilot and data subcarriers within the interference band are contaminated. One could adopt a Bayesian estimator as

$$\{\hat{\mathbf{H}}, \hat{\mathbf{s}}, \hat{\mathbf{u}}, \hat{\tau}_1\} = \arg \max_{\{\mathbf{H}, \mathbf{s}, \mathbf{u}, \tau_1\}} f(\mathbf{z}|\mathbf{H}, \mathbf{s}, \mathbf{u}, \tau_1) f(\mathbf{H}) f(\mathbf{s}) f(\mathbf{u}) f(\tau_1). \quad (23)$$

However, the computational complexity of solving (23) directly is prohibitive.

To reduce the computational load in solving (23), we propose an iterative receiver structure for interference detection/estimation, channel estimation and information symbol detection, as shown in Fig. 2. Relative to the iterative/progressive receiver for a time-varying channel of [40], the proposed structure includes several new modules for interference detection and estimation. It is thus a more general framework than the one in [40] in which only the self-interference in the form of ICI is addressed.

The modules of the proposed receiver are briefly described as follows.

- *Preprocessing* The main Doppler effect and residual Doppler shift are removed via the resampling operation and the Doppler compensation, respectively [7]. Set the initial value of the iteration index $I = 0$.

- *Generalized Log-Likelihood Ratio Test (GLRT) Interference Estimation/Detection* A GLRT detector is used to detect the interference. During this process, maximum likelihood (ML) estimates of the interference $\mathbf{u}$ and the delay τ_1 are obtained. Comparison of the GLRT statistics with a predetermined threshold yields a decision about the presence of the interference.
- *Interference subtraction* If interference exists, the contributions of interference will be estimated and subtracted from the measurements.
- *Channel estimation* Based on both pilot symbols and the soft decisions on information symbols fed back from the decoder, channel estimation methods from a conventional OFDM receiver can be applied. We herein adopt the sparse channel estimation from [13].
- *Channel equalization* Multiple equalizers, such as zero-forcing (ZF), minimum mean square error (MMSE), and Markov-Chain Monte Carlo (MCMC) can be used. In this paper, we adopt the linear MMSE equalizer [41]. The *a priori* information about the data symbols comes from the nonbinary LDPC decoder in the preceding iteration, and the extrinsic information from the equalizer will be used as prior information to the LDPC decoder.
- *Nonbinary LDPC decoding* Based on the extrinsic information from the LMMSE equalizer, the nonbinary LDPC decoder [42] generates the decoded information symbols and the soft decisions about the information symbols for both channel estimation and LMMSE equalization in the next iteration.
- *Noise variance update* Due to the iterative estimation of the interference, the channel mixing matrix and information symbols, the noise variance is kept updated within each iteration.

Once the information symbols have been successfully decoded, i.e., the parity check conditions of the nonbinary LDPC decoder are satisfied, or the number of iterations reaches a predetermined threshold $I_{\max}$, the iterative loop will stop.

We start with the initialization of the proposed iterative receiver in Section IV-A. A detailed description of GLRT interference detection will be presented in Section IV-B, and sparse channel estimation, LMMSE equalization, and nonbinary LDPC decoding will be discussed in Section IV-C. Estimation for the variance of noise within and outside of the interference band is described in Section IV-D.

A. Initialization

In the OFDM signal design, null subcarriers are usually inserted into the signal band to estimate the carrier frequency offset [7] and the variance of the ambient noise [43]. To initialize the iterative interference cancellation in Fig. 2, the following operations are carried out.

- 1) Based on the frequency measurements at null subcarriers, an linear interpolation is performed to estimate the variance of interference within the signal band.
- 2) Assume that the interference components at all the frequency subcarriers are independent. A prewhitening operation is performed for the frequency measurements [44].

- 3) With the prewhitened frequency measurements, the channel estimation is carried out based on the frequency measurements at pilot subcarriers, which is followed by the ICI equalization and nonbinary LDPC decoder to estimate the information symbols and to recover the information bits.
- 4) If the parity check conditions of the nonbinary LDPC decoder cannot be satisfied, the estimated information symbols are fed into the GLRT interference detector to initialize the iterative interference mitigation.

B. Interference Detection and Estimation

Let us first assume that both channel and symbol estimates ($\hat{\mathbf{H}}$ and $\hat{\mathbf{s}}$) are available. Within the interference band $\mathcal{B}_I$, there are $M_I = \lceil B_I T \rceil$ subcarriers contaminated. Denote the set of subcarriers within the interference band as $\{i_1, \dots, i_{M_I}\}$. Define a selector matrix $\mathbf{\Theta}$ of size $M_I \times K$ with unit entry at the (m, i_m) th position ($m = 1, \dots, M_I$) and zeros elsewhere. The selection matrix $\mathbf{\Theta}$ is determined based on the prior knowledge of the interference frequency band.

The relevant measurements within the interference frequency band are contained in

$$\bar{\mathbf{z}} = \mathbf{\Theta}(\mathbf{z} - \hat{\mathbf{H}}\hat{\mathbf{s}}) = \mathbf{B}(\tau_I)\mathbf{u} + \bar{\mathbf{w}} \quad (24)$$

where

$$\mathbf{B}(\tau_I) := \mathbf{\Theta}\mathbf{\Lambda}(\tau_I)\mathbf{\Gamma}_I, \quad \bar{\mathbf{w}} = \mathbf{\Theta}\mathbf{w} + \mathbf{\Theta}(\hat{\mathbf{H}}\hat{\mathbf{s}} - \mathbf{H}\mathbf{s}). \quad (25)$$

Here, $\bar{\mathbf{w}}$ denotes the equivalent additive noise within the frequency band, which consists of the ambient noise and the residual noise due to imperfect channel and information symbol estimates.

Assume that the noise samples are independent and follow a complex Gaussian distribution $\mathcal{CN}(0, \sigma_{\mathcal{B}_I}^2 \mathbf{I}_{M_I})$, where $\sigma_{\mathcal{B}_I}^2$ denotes the noise variance. The likelihood function of the interference component $\bar{\mathbf{z}}$ in the presence of interference is

$$f(\bar{\mathbf{z}}|\tau_I, \mathbf{u}) \propto \exp \left[-\frac{1}{\sigma_{\mathcal{B}_I}^2} \|\bar{\mathbf{z}} - \mathbf{B}(\tau_I)\mathbf{u}\|^2 \right]. \quad (26)$$

Let $\mathcal{H}_0$ and $\mathcal{H}_1$ denote the absence and presence of interference, respectively. To detect the presence of interference in one particular OFDM block, we define the GLRT statistic

$$\begin{aligned} L(\bar{\mathbf{z}}) &= \max_{\{\tau_I, \mathbf{u}\}} \ln \frac{f(\bar{\mathbf{z}}|\tau_I, \mathbf{u}, \mathcal{H}_1)}{f(\bar{\mathbf{z}}|\mathcal{H}_0)} \\ &= \max_{\{\tau_I, \mathbf{u}\}} \ln \frac{\exp \left[-\frac{\|\bar{\mathbf{z}} - \mathbf{B}(\tau_I)\mathbf{u}\|^2}{\sigma_{\mathcal{B}_I}^2} \right]}{\exp \left[-\frac{\|\bar{\mathbf{z}}\|^2}{\sigma_{\mathcal{B}_I}^2} \right]} \\ &= \max_{\{\tau_I, \mathbf{u}\}} \frac{1}{\sigma_{\mathcal{B}_I}^2} \left[\bar{\mathbf{z}}^H \mathbf{B}(\tau_I)\mathbf{u} + \mathbf{u}^H \mathbf{B}^H(\tau_I)\bar{\mathbf{z}} \right. \\ &\quad \left. - \mathbf{u}^H \mathbf{B}^H(\tau_I)\mathbf{B}(\tau_I)\mathbf{u} \right] \geq \Gamma_{\text{th}} \end{aligned} \quad (27)$$

where Γ_{th} is a predetermined threshold.

Define an objective function to be maximized over $\{\tau_I, \mathbf{u}\}$

$$J \triangleq \bar{\mathbf{z}}^H \mathbf{B}(\tau_I)\mathbf{u} + \mathbf{u}^H \mathbf{B}^H(\tau_I)\bar{\mathbf{z}} - \mathbf{u}^H \mathbf{B}^H(\tau_I)\mathbf{B}(\tau_I)\mathbf{u}. \quad (28)$$

Setting the derivative $\nabla J_{\mathbf{u}}$ to zero yields the optimal estimate of $\mathbf{u}$ as

$$\hat{\mathbf{u}} = [\mathbf{B}^H(\tau_I)\mathbf{B}(\tau_I)]^{-1} \mathbf{B}^H(\tau_I)\bar{\mathbf{z}} \quad (29)$$

Substituting $\hat{\mathbf{u}}$ into (28), the estimate of τ_I is obtained as

$$\hat{\tau}_I = \arg \max_{\tau_I} \bar{\mathbf{z}}^H \mathbf{B}(\tau_I) [\mathbf{B}^H(\tau_I)\mathbf{B}(\tau_I)]^{-1} \mathbf{B}(\tau_I)^H \bar{\mathbf{z}} \quad (30)$$

which can be solved using the grid search.

Based on the estimated parameters $\{\hat{\mathbf{u}}, \hat{\tau}_I\}$, the test statistic is evaluated as

$$L(\bar{\mathbf{z}}) = \frac{1}{\hat{\sigma}_{\mathcal{B}_I}^2} \hat{\mathbf{u}}^H \mathbf{B}^H(\hat{\tau}_I)\mathbf{B}(\hat{\tau}_I)\hat{\mathbf{u}} \geq \Gamma_{\text{th}} \quad (31)$$

where the estimate $\hat{\sigma}_{\mathcal{B}_I}^2$ is obtained following the procedure to be discussed in Section IV-D. The test statistic $L(\bar{\mathbf{z}})$ is actually the ratio of the energy of the estimated interference to the energy of the equivalent noise.

1) *Determination of the GLRT Threshold:* Note that $L(\bar{\mathbf{z}}|\mathcal{H}_1)$ follows a central χ^2 distribution $f(L(\bar{\mathbf{z}}); \kappa)$, and $L(\bar{\mathbf{z}}|\mathcal{H}_0)$ follows a noncentral χ^2 distribution $f(L(\bar{\mathbf{z}}); \kappa, \lambda)$, both with degrees of freedom $\kappa = M_I$. Denote the energy of the interference by $E_I = \mathbf{u}^H \mathbf{B}^H(\tau_I)\mathbf{B}(\tau_I)\mathbf{u}$. The noncentrality parameter is $\lambda = \frac{E_I}{\sigma_{\mathcal{B}_I}^2}$. The test threshold Γ_{th} then can be determined based on the predetermined probability of false alarm P_{FA} or the probability of detection P_D

$$\begin{aligned} P_{\text{FA}} &= \int_{\Gamma_{\text{th}}}^{\infty} f(L(\bar{\mathbf{z}}|\mathcal{H}_0); \kappa) dL(\bar{\mathbf{z}}|\mathcal{H}_0) = 1 - P \left(\frac{\Gamma_{\text{th}}}{2}, \frac{\kappa}{2} \right) \\ P_D &= \int_{\Gamma_{\text{th}}}^{\infty} f(L(\bar{\mathbf{z}}|\mathcal{H}_1); \kappa, \lambda) dL(\bar{\mathbf{z}}|\mathcal{H}_1) = Q_{\frac{\kappa}{2}} \left(\sqrt{\lambda}, \sqrt{\Gamma_{\text{th}}} \right) \end{aligned} \quad (32)$$

where $P(a, b)$ is the incomplete Gamma function, and $Q_M(a, b)$ is the Marcum Q-function [45].

2) *Computational Complexity Reduction:* One straightforward approach to solving (30) is an exhaustive search over all possible values of τ_I . To reduce the computational load, we define Ψ as a $K \times K$ diagonal matrix with the diagonal elements formed by $\mathbf{\Theta}^H \bar{\mathbf{z}}$, and a $K \times 1$ column vector $\phi(\tau_I)$ containing the diagonal entries of the matrix $\mathbf{\Lambda}(\tau_I)$. It is easy to verify that $\mathbf{B}(\tau_I)^H \mathbf{B}(\tau_I) = \mathbf{\Gamma}_I^H \mathbf{\Theta}^H \mathbf{\Theta} \mathbf{\Gamma}_I$. Hence, we can rewrite (30) as

$$\hat{\tau}_I = \arg \max_{\tau_I} \underbrace{\phi^H(\tau_I) \Psi^H \mathbf{\Gamma}_I \left(\mathbf{\Gamma}_I^H \mathbf{\Theta}^H \mathbf{\Theta} \mathbf{\Gamma}_I \right)^{-1} \mathbf{\Gamma}_I^H \Psi \phi(\tau_I)}_{:= \mathbf{J}}. \quad (34)$$

Given a predetermined searching step and searching range of τ_I , a partial fast Fourier transform (FFT) matrix Φ can be formed by stacking the column vectors $\phi(\tau_I)$ at all possible values of τ_I . The optimal τ_I can be obtained by finding the maximum diagonal element of $\Phi^H \mathbf{J} \Phi$, where $\mathbf{J}$ is defined in (34). The matrix $\mathbf{J}$ can be pre-computed given the prior knowledge of the time duration and frequency band of the interference. Using FFT to compute $\Phi^H \mathbf{J}$, finding the diagonal elements of $\Phi^H \mathbf{J} \Phi$ only requires a complexity on the order of $K^2(\log_2(K) + 1)$.

C. Channel Estimation, Equalization and Nonbinary LDPC Decoding

If the presence of interference is declared from the GLRT detector (either true detection or false alarm), the desired OFDM component can be obtained by subtracting the estimated interference from the received signal

$$\tilde{\mathbf{z}} = \mathbf{z} - \Lambda(\hat{\tau}_1)\mathbf{\Gamma}_1\hat{\mathbf{u}} = \mathbf{H}\mathbf{s} + \tilde{\mathbf{w}} \quad (35)$$

where $\tilde{\mathbf{w}}$ denotes the equivalent noise which consists of the ambient noise and the residual interference

$$\tilde{\mathbf{w}} = \mathbf{w} + [\Lambda(\tau_1)\mathbf{\Gamma}_1\mathbf{u} - \Lambda(\hat{\tau}_1)\mathbf{\Gamma}_1\hat{\mathbf{u}}]. \quad (36)$$

If no interference is detected (either absence of interference or missed detection), we simply set $\hat{\mathbf{u}} = 0$ in (35). The following processing will be carried out based on $\tilde{\mathbf{z}}$.

Based on the observation vector $\tilde{\mathbf{z}}$ and the symbol vector $\hat{\mathbf{s}}$, the sparse channel estimator in [13] is adopted in this paper to estimate the channel matrix $\mathbf{H}$, where the banded assumption in (9) is used to reduce computational complexity.

Channel equalization is then carried out according to the maximum *a posteriori* (MAP) or the minimum mean square error (MMSE) criteria. In this paper, the linear MMSE estimator is adopted as the performance benchmark [46]

$$\hat{\mathbf{s}} = \bar{\mathbf{s}} + \text{Cov}(\mathbf{s}, \tilde{\mathbf{z}})\text{Cov}(\tilde{\mathbf{z}}, \tilde{\mathbf{z}})^{-1}(\tilde{\mathbf{z}} - \hat{\mathbf{H}}\bar{\mathbf{s}}) \quad (37)$$

where $\bar{\mathbf{s}}$ is the prior mean vector, and $\text{Cov}(\mathbf{a}, \mathbf{b})$ denotes the covariance matrix of vectors $\mathbf{a}$ and $\mathbf{b}$. Since a nonbinary LDPC decoder [42] is involved in the receiver, the prior information about the data symbols from the decoder should be excluded from the LMMSE estimate by setting the prior mean and variance of $s[m]$ to 0 and E_s , respectively

$$\begin{aligned} \hat{s}[m] &= \text{Cov}(s[m], \tilde{\mathbf{z}})\text{Cov}(\tilde{\mathbf{z}}, \tilde{\mathbf{z}})^{-1}(\tilde{\mathbf{z}} - \hat{\mathbf{H}}\bar{\mathbf{s}} + \hat{\mathbf{h}}_m\bar{s}[m]), \\ &= \mathbf{f}_m^H(\tilde{\mathbf{z}} - \hat{\mathbf{H}}\bar{\mathbf{s}} + \hat{\mathbf{h}}_m\bar{s}[m]) \end{aligned} \quad (38)$$

with

$$\mathbf{f}_m \triangleq E_s \left[\sigma_s^2 \hat{\mathbf{H}}\hat{\mathbf{H}}^H + \hat{\sigma}_w^2 \mathbf{I}_K + (E_s - \sigma_s^2)\hat{\mathbf{h}}_m\hat{\mathbf{h}}_m^H \right]^{-1} \hat{\mathbf{h}}_m \quad (39)$$

where $\bar{s}[m]$ and σ_s^2 are the prior mean and variance of information symbol $s[m]$, respectively, $\hat{\mathbf{h}}_m$ is the m th column of channel matrix $\hat{\mathbf{H}}$, and $\hat{\sigma}_w^2$ is the variance of the equivalent noise which will be estimated in Section IV-D.

After inputting the LMMSE estimate of information symbols into the nonbinary LDPC decoder, both hard and soft decisions on the information symbols can be obtained, these being fed back for interference detection, channel estimation and equalization in the next iteration. A detailed description of the nonbinary LDPC decoder can be found in [42].

D. Noise Variance Estimation

Due to the partial-band property of the interference, the noise variance σ_w^2 should be estimated individually for noise within and outside of the interference band $\mathcal{B}_1$ based on the frequency measurements at null subcarriers. Based on the estimates of the channel matrix and transmitted symbols, the variance of the

TABLE I
OFDM PARAMETERS IN SIMULATION AND SPACE08 EXPERIMENT

center frequency	f_c	13 kHz
bandwidth	B	9.77 kHz
# of subcarriers	K	1024
symbol duration	T	104.86 ms
frequency spacing	$\Delta f \triangleq 1/T$	9.54 Hz
guard interval	T_g	24.6 ms

equivalent noise outside of the interference band, which consists of the ambient noise and the residual ICI due to the banded assumption of the channel matrix, can be estimated as

$$\hat{\sigma}_{\mathcal{B}_1}^2 = \mathbb{E}_{\{m \in \mathcal{S}_N, f_m \notin \mathcal{B}_1\}} \left[\left\| z[m] - \sum_{k=m-D}^{m+D} \hat{H}[m, k] \hat{s}[k] \right\|^2 \right]. \quad (40)$$

For the equivalent noise within the interference band, which consists of the ambient noise, the residual ICI and the residual interference, the noise variance can be estimated as

$$\begin{aligned} \hat{\sigma}_{\mathcal{B}_1}^2 &= \mathbb{E}_{\{m \in \mathcal{S}_N, f_m \in \mathcal{B}_1\}} \left[\left\| z[m] - \sum_{k=m-D}^{m+D} \hat{H}[m, k] \hat{s}[k] \right. \right. \\ &\quad \left. \left. - \sum_{l=-\frac{N_1}{2}}^{\frac{N_1}{2}-1} \Lambda(\hat{\tau}_1)[m, m] \mathbf{\Gamma}_1[m, l] \hat{u}[l] \right\|^2 \right]. \end{aligned} \quad (41)$$

The estimated variance is then used for interference detection and information symbol estimation.

V. SIMULATION RESULTS

We simulate a sparse channel consisting of 10 discrete paths, where the interarrival time follows an exponential distribution with a mean of 1 ms. The amplitudes are Rayleigh distributed with average power decreasing exponentially with the delay, where the difference between the beginning and the end of the guard time is 20 dB.

The ZP-OFDM signal parameters are listed in Table I, which are tailored according to the setting of one field experiment in Section VII. Out of 1024 subcarriers, 96 are null subcarriers with 24 on each edge for band protection and 48 distributed evenly in the middle, 256 are pilot subcarriers for channel estimation, and the remaining 672 are data subcarriers for information delivering. The data symbols are encoded with a rate-1/2 nonbinary LDPC code [42] and modulated with a 16-QAM constellation, which leads to a data rate

$$R = \frac{1}{2} \frac{|\mathcal{S}_D|}{T + T_g} \log_2 16 = 10.4 \text{ kb/s} \quad (42)$$

where $\mathcal{S}_D$ denotes the set of data subcarriers.

The interference is generated by passing white Gaussian noise of a given duration through a bandpass filter according to the following parameters: center frequency $f_{1c} = 15$ kHz, bandwidth $B_1 = 2.4$ kHz and time duration $T_1 = 26.2$ ms. The delay of the interference relative to start of each OFDM block is uniformly distributed within $[0, T + T_g - T_1]$.

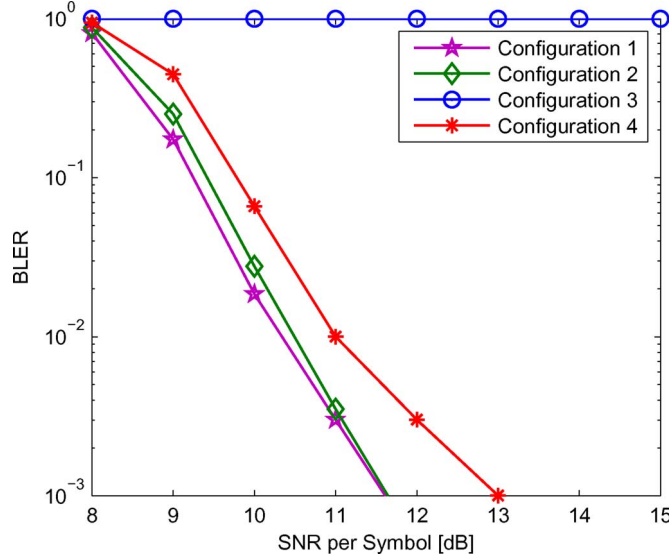


Fig. 3. BLER performance of several receivers in the time-invariant scenario, $SIR = 0$ dB.

Throughout this paper, the signal-to-noise ratio (SNR) and signal-to-interference ratio (SIR) are defined as

$$SNR = \frac{P_s}{N_0}, \quad SIR = \frac{P_s}{P_I} \quad (43)$$

where P_s denotes the average power of OFDM frequency measurements within the useful signal band, P_I denotes the average power of interference frequency components within the interference frequency band, and N_0 denotes the variance of the additive noise in frequency domain.

Four configurations are introduced to compare the performance of the proposed receiver with the receiver which does not perform interference cancellation.

- *Configuration 1*: the interference-free environment; the conventional receiver is adopted;
- *Configuration 2*: the interference-free environment; the proposed receiver is adopted;
- *Configuration 3*: the interference-present environment; the conventional receiver is adopted;
- *Configuration 4*: the interference-present environment; the proposed receiver is adopted.

The block-error-rate (BLER) performance will be used as the performance metric. When applying the proposed receiver in configurations 2 and 4, we assume that the interference has been detected in each OFDM block such that interference cancellation is always performed.

A. Time-Invariant UWA Channels

To examine the performance of the proposed receiver in a time-invariant channel, the Doppler rate of each path is set zero. Averaged over 2000 Monte Carlo runs, Fig. 3 demonstrates the BLER performance of the four configurations with $SIR = 0$ dB. In the presence of interference, the receiver without interference cancellation does not work, while the proposed receiver can approach the performance of the conventional receiver in the absence of interference. Meanwhile, in the scenario without

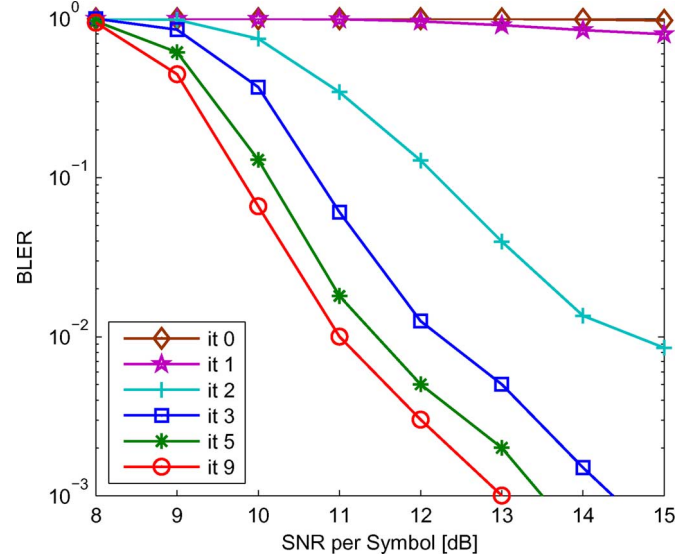


Fig. 4. BLER performance improvement against iterations of the proposed receiver in the time-invariant scenario $SIR = 0$ dB.

interference, there is only slight performance degradation of the proposed receiver relative to the conventional receiver.

Fig. 4 shows the performance improvement of the proposed receiver with different number of iterations. One can observe a large performance gap between the receiver without iteration, i.e., the receiver which performs interference suppression via noise prewhitening as described in Section IV-A, and the proposed receiver which iteratively mitigates the interference through interference estimation and subtraction.

B. Time-Varying UWA Channels

To explore the receiver performance in a time-varying channel, the Doppler rate of each path is drawn *independently* from a zero-mean uniform distribution with the standard deviation σ_v m/s. To improve the channel estimation performance in the presence of ICI, we convert 96 data subcarriers as extra pilot subcarriers, which are grouped with the null subcarriers and original pilot subcarriers into clusters [13]. The new data rate R is recomputed as

$$R = \frac{|S_D| - 96}{T + T_g} \log_2 16 = 7.4 \text{ kb/s}. \quad (44)$$

Fig. 5 shows the performance of the four configurations in the scenario with a mild Doppler spread $\sigma_v = 0.1 \frac{\text{m}}{\text{s}}$ and a significant Doppler spread $\sigma_v = 0.2 \frac{\text{m}}{\text{s}}$, respectively. Resampling operation is not performed (i.e., $\hat{a} = 0$). A banded assumption of the channel matrix is adopted with the ICI depth $D = 1$. We can observe that in the presence of interference, the performance of the proposed receiver still can approach that of the conventional receiver in the scenario without interference, while the conventional receiver without interference cancellation does not work.

C. Performance of the Proposed Receiver With Different SIRs

By varying the SIR level, performances of the proposed receiver in both time-invariant and time-varying scenarios are shown in Fig. 6. One can observe that when the SIR is

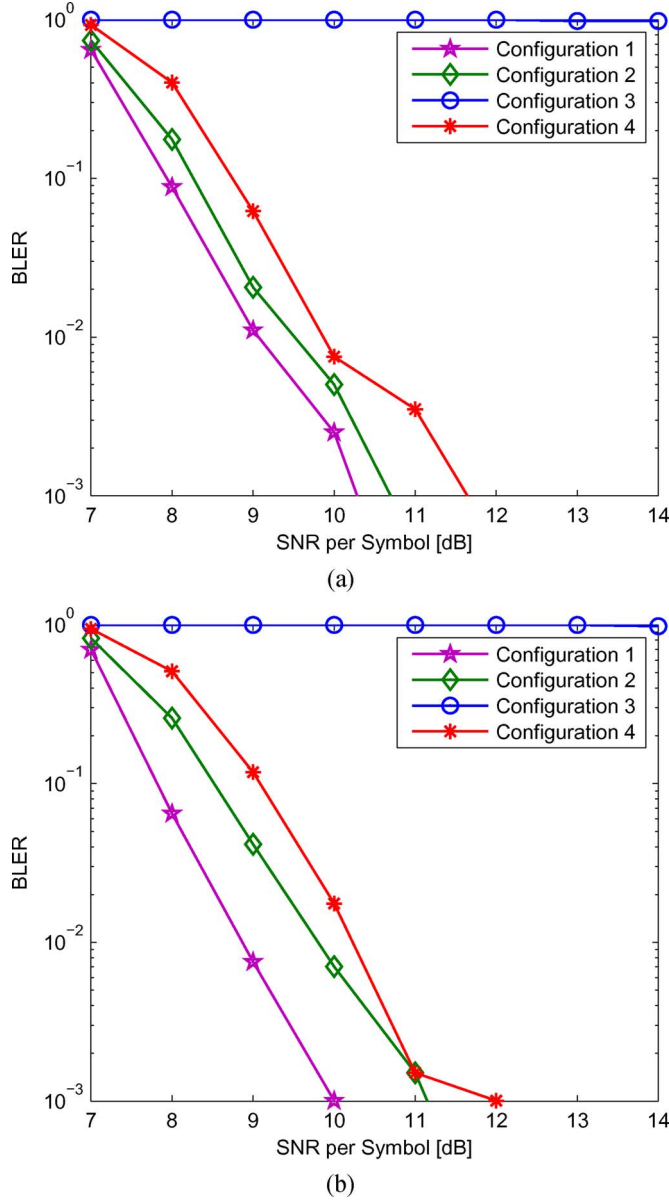


Fig. 5. BLER performance of receivers in the time-varying channels, $D = 1$, SIR = 0 dB. (a) $\sigma_v = 0.1$ m/s. (b) $\sigma_v = 0.2$ m/s.

very small, a BLER error floor shows up, and that as the SIR increases, the decoding performance of the proposed receiver gradually converges to the decoding performance in the scenario without interference.

D. Interference Detection and Estimation

Define E_I as the interference energy, and N_0 as the power of ambient noise. The ROC curve of the GLRT detector with different $\frac{E_I}{N_0}$ and time-bandwidth product of interference is shown in Fig. 7. We can observe that with an identical $\frac{E_I}{N_0}$, the larger time-bandwidth product of the interference, the more difficult for it to be detected, because it appears more like the ambient noise.

VI. EXPERIMENTAL RESULTS: AUTECH

This experiment was held in March 2010 at the Atlantic Undersea Test and Evaluation Center (AUTC) around Andros Is-

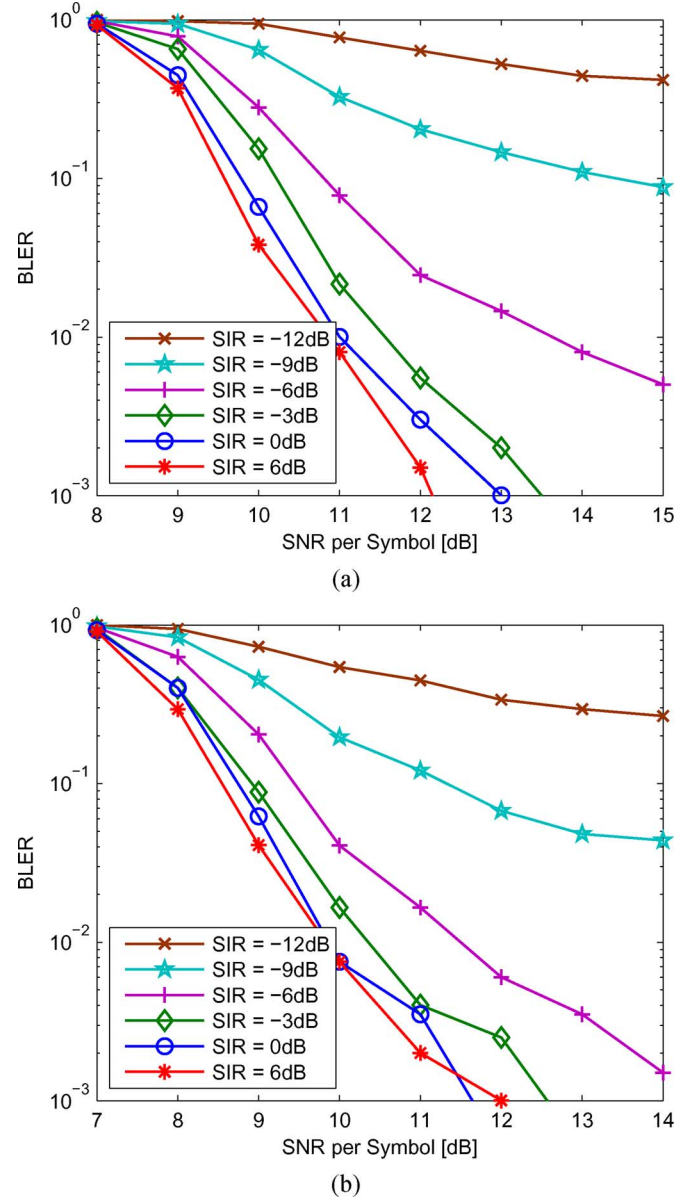


Fig. 6. BLER performance of the proposed receiver at different SIR levels. (a) Time-invariant scenario $D = 0$. (b) Time-varying scenario $\sigma_v = 0.1$ m/s, $D = 1$.

land near the Tongue of the Ocean, Bahamas [2], [3]. There is one transmitter and 95 receiving hydrophones located in an area of the size $30 \text{ km} \times 30 \text{ km}$. Only several receiving hydrophones are used in this paper. The receivers are at least 4 km apart from each other. Depths of the transmitter and the receivers vary from 1.5 to 2 km.

Parameters of the transmitted OFDM signal are shown in Table II. The total 860 subcarriers comprise a random distribution of three pilot subcarrier patterns and three data subcarrier patterns

- pilot patterns: $\{[N P N]_{(50)}, [P P]_{(16)}, [P P P]_{(114)}\}$
- data patterns: $\{[D]_{(48)}, [D D]_{(105)}, [D D D]_{(26)}\}$,

where N, P, and D denote null subcarrier, pilot subcarrier, and data subcarrier, respectively, and the subindex denotes the number of different patterns, which results in 424 pilot subcarriers and 336 data subcarriers in total. Again, a rate-1/2

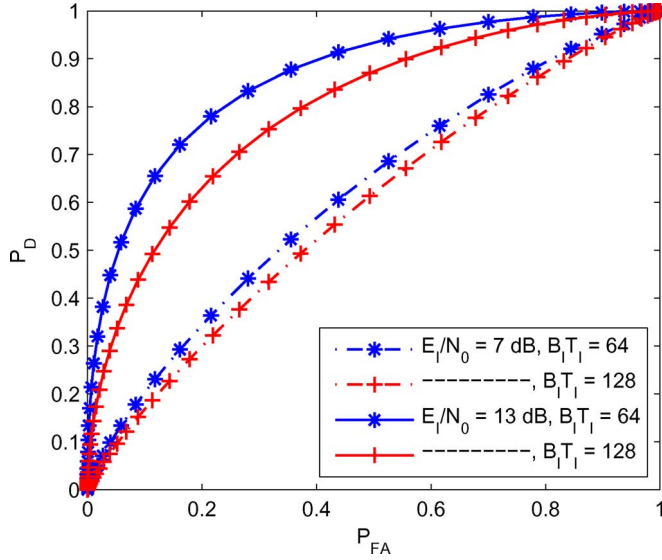
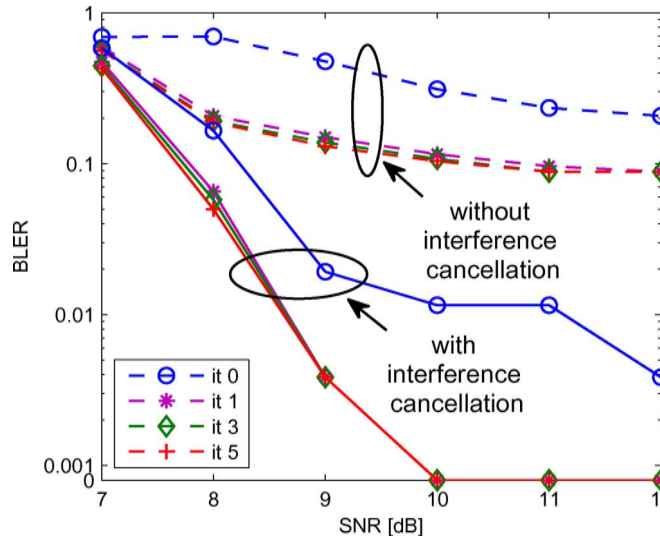


Fig. 7. ROC of the GLRT interference detector.

TABLE II
PARAMETERS OF ZP-OFDM IN AUTEC10 EXPERIMENT

center frequency	f_c	11 kHz
bandwidth	B	5 kHz
# of subcarriers	K	860
symbol duration	T	170.7 ms
subcarrier spacing	$\Delta f \triangleq 1/T$	5.86 Hz
guard interval	T_g	250 ms
# of blocks per transmission	N_b	10

Fig. 8. Block error rate of 13 transmissions with/without interference cancellation (IC) versus different levels of added noise $I_{\max} = 7$.

nonbinary LDPC code and a QPSK constellation are used, leading to a data rate

$$R = \frac{1}{2} \frac{336 \times \log_2 4}{0.1707 + 0.25} = 798.7 \text{ bits/s} \quad (45)$$

or

$$\frac{3R}{8} = 299.5 \text{ Bytes/(3 s)} \quad (46)$$

where Bytes/(3 s) is the unit used in the current AUTEC system.

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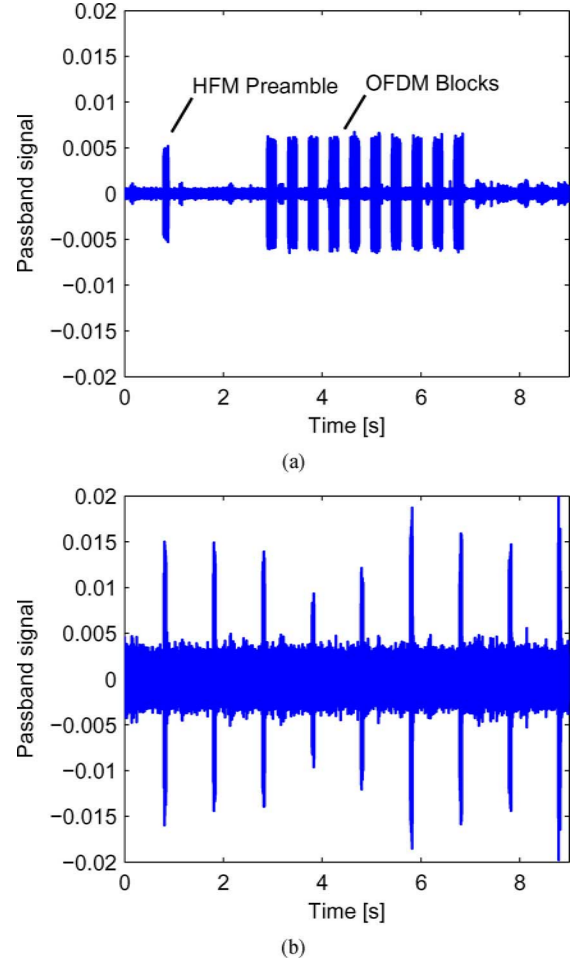
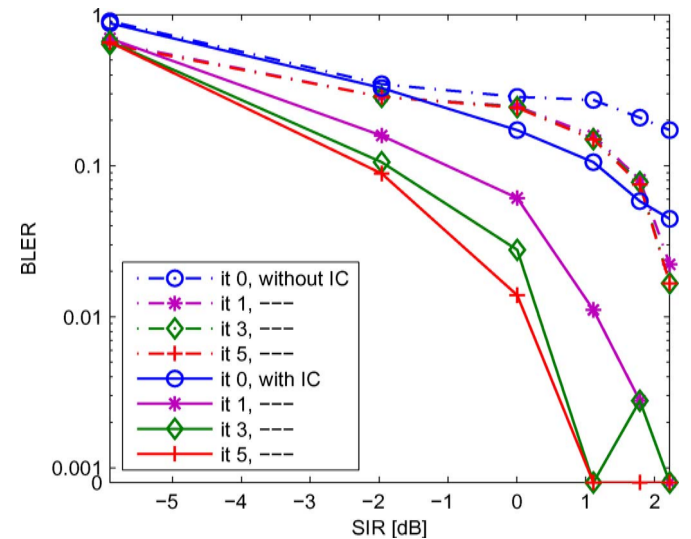


Fig. 9. Sample of the received signal at two hydrophones after bandpass filtering. The two data sets are added together with different weights to generate semiexperimental data sets at different SIR levels. (a) Hydrophone (a). (b) Hydrophone (b).

Fig. 10. Block error rate of 18 transmissions with/without interference cancellation (IC) versus different signal to interference power ratio, $I_{\max} = 7$, $\text{SNR} \approx 7.9 \text{ dB}$.

A hyperbolic frequency modulated (HFM) waveform with time duration 100 ms and bandwidth 5 kHz was used as the preamble to probe the channel.

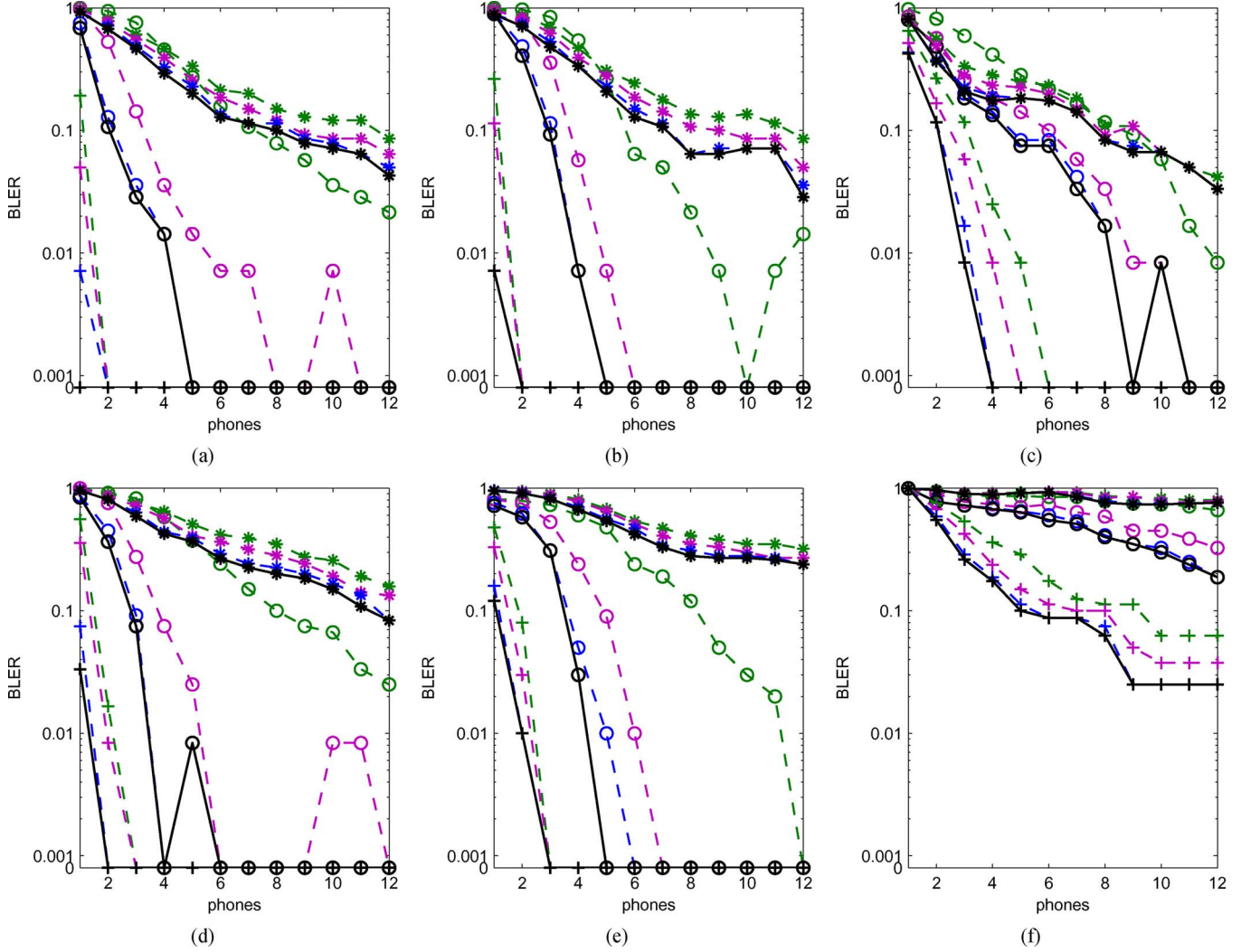


Fig. 11. BLER performance comparison of receiver with/without interference cancellation, 16 QAM, $D = 3$. Marker *: receiver without interference cancellation; marker o: proposed interference cancellation receiver; marker +: original BLER without adding interference; dashed line: [0 1 5 9]th iteration, solid line: 10th iteration. (a) S1, SIR = -3.0 dB, Julian date: 297. (b) S3, SIR = -2.8 dB, Julian date: 297. (c) S5, SIR = 0.4 dB, Julian date: 297. (d) S1, SIR = -4.2 dB, Julian date: 300. (e) S3, SIR = -3.5 dB, Julian date: 300. (f) S5, SIR = -2.1 dB, Julian date: 300.

There are two transmission trials of an identical data set, with the first trial and the second trial including 18 transmissions and 22 transmissions, respectively. During the experiment, sonar operations were carried out by others simultaneously. At sensors close to the sonar operation, the received signal of all the transmissions in the first trial and the first portion of transmissions in the second trial were contaminated. Based on the received signal waveform in Fig. 1, the receiver can infer that $T_1 \approx 45$ ms and the frequency band $[11, 15]$ kHz, and hence sets $f_{1c} \approx 13$ kHz, the interfering bandwidth $B_I \approx 2$ kHz.

In the sequel, the proposed interference cancellation receiver will be tested with the experimental data in two different cases. Note that as opposed to the simulations, only parts of the OFDM blocks contain interference; we thus start the interference detection module in the receiver to decide the necessity of interference cancellation.

A. Test Case 1

In this test case, we investigate the interference cancellation performance of the receiver at different SNR levels. Thirteen

transmissions with the SIR level ranging from 0 to 9 dB from the second trial are used for this test. With the received interference-contaminated waveform shown in Fig. 1, white Gaussian noise is added to the received signal to generate several semiexperimental data sets with different SNRs.

The decoding performance of the semiexperimental data sets with and without interference mitigation is shown in Fig. 8. One can observe that the receiver with interference cancellation outperforms the receiver without interference cancellation, and the performance gap between the iterative interference cancellation method and the noise-prewhitening method is large.

B. Test Case 2

In this test case, we investigate the interference cancellation performance of the receiver at different SIR levels. By modifying the weighting coefficients, the semiexperimental data sets of different SIRs are constructed through adding the received signal of pure interference with one received OFDM signal which is not contaminated by interference. The received signal at two hydrophones are shown in Fig. 9, where the

signal at hydrophone (a) is interference-free, while the signal at hydrophone (b) is almost purely interference with a very weak useful signal masked by ambient noise.

Fig. 10 shows the decoding performance of the semiexperimental data sets with and without interference cancellation. Again, we can observe that the receiver with interference cancellation outperforms that without interference cancellation considerably, and the performance improvement brought by the iterative interference cancellation is significant.

VII. EXPERIMENTAL RESULTS: SPACE08

This experiment was held off the coast of Martha's Vineyard, MA, from Oct. 14 to Nov. 1, 2008. The water depth was about 15 m. Among all the six receivers, we only consider the data collected by three receivers, labeled as S1, S3, S5, which were 60, 200, and 1000 m away from the transmitter, as in [13]. Each receiver array consists of twelve hydrophones. During the experiment, two cycles of large waves showed up, one around Julian date 297 and the other around Julian date 300. The latter storm was more severe. We thus consider the data recorded from these two days to verify performance of the proposed method in the time-varying scenario. For each day, there are twelve recorded files, each consisting of twenty OFDM blocks. We test all the files on Julian date 297. On Julian date 300, five files recorded in the afternoon were severely distorted; we thus only consider the seven files recorded in the morning and evening. Parameter settings of this experiment are summarized in Table I.

Note that there was no external interference during this experiment. We extract the interference from the recorded sonar waveform in the AUTE10 experiment at hydrophone (b) as shown in Fig. 9, and artificially add each sonar waveform to each OFDM block to create a semi-experimental data set with interference contamination. The delay of the added sonar waveform relative to the OFDM block follows a uniform distribution within the interval $[0, T + T_g - T_1]$. To estimate the channel accurately, 96 data subcarriers are used as extra pilot subcarriers, as discussed in Section V-B. The data rate as stated in (45) is achieved. Again, different from the simulations, the receiver incorporates the interference detection module to determine the necessity of interference cancellation.

The decoding performance of the proposed receiver and the conventional receiver without interference cancellation, and the decoding performance of the signal without adding the interference are depicted in Fig. 11. One can find that relative to the decoding performance of the original received signal, significant performance degradation of the conventional receiver is incurred by introducing the interference. Meanwhile, the proposed receiver can effectively eliminate the interference after a number of iterations.

VIII. CONCLUSION

In this paper, we developed an interference mitigation receiver for OFDM in underwater acoustic communications. Assuming prior knowledge of frequency band and time duration of the interference, the interference waveform was parameterized by a limited number of parameters, where the number of parameters is equal to the time-bandwidth product. Incorporating the

interference into the OFDM system model while accounting for the unknown delay offset of the interference waveform relative to the OFDM block, an OFDM receiver with explicit interference mitigation was developed. A GLRT was used for interference detection, and interference estimation and data decoding were carried out iteratively. Both simulation and experimental results demonstrated that the iterative interference mitigation considerably outperforms the noniterative noise-prewhitening approach which treats interference as ambient noise. Particularly in our experimental data sets, the performance of the proposed receiver in the presence of interference can approach that of the traditional receiver in the scenario without interference, which shows that the proposed receiver has robust performance against interference.

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